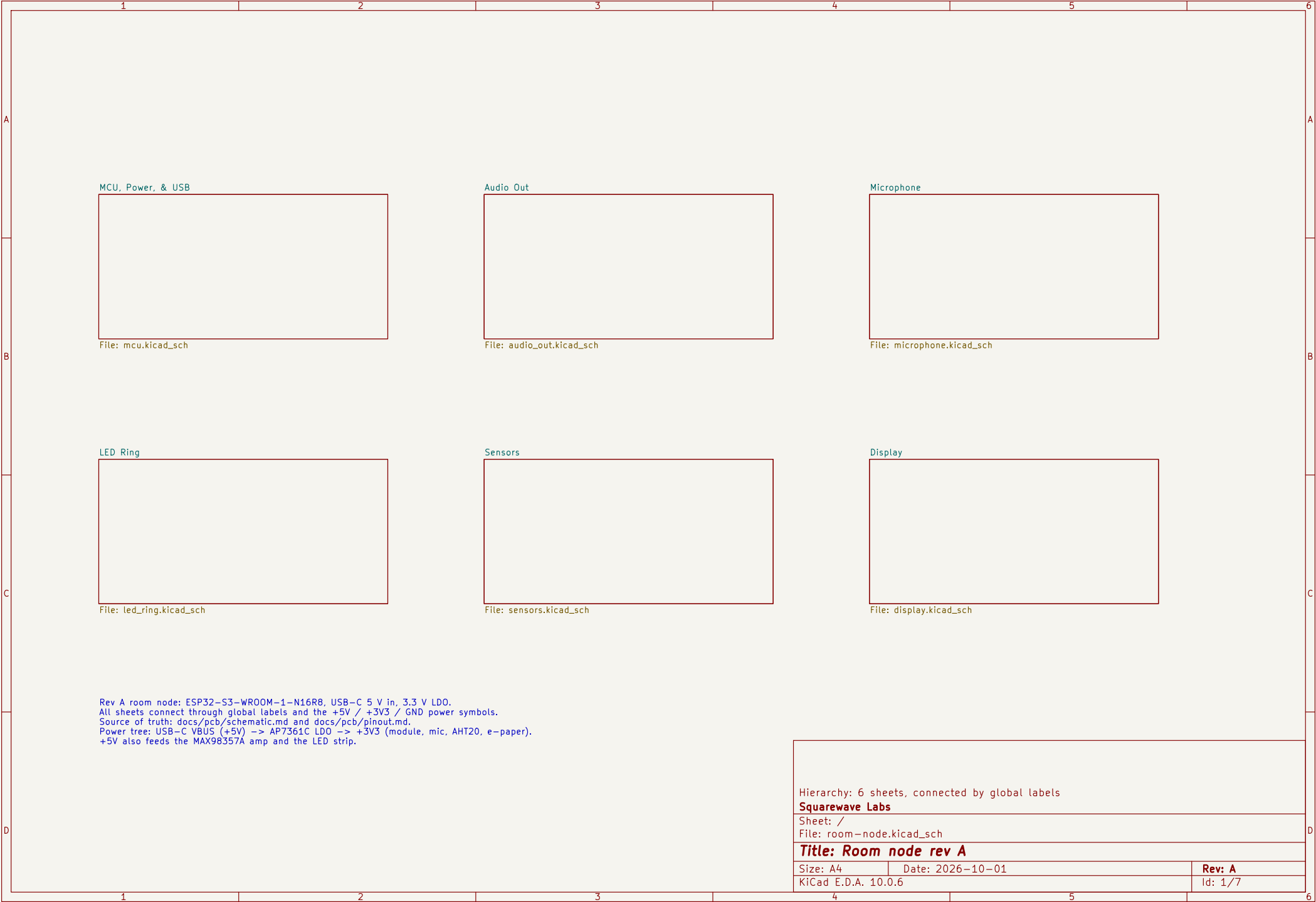




MCU, Power, & USB

File: mcu.kicad_sch

Audio Out

File: audio_out.kicad_sch

Microphone

File: microphone.kicad_sch

LED Ring

File: led_ring.kicad_sch

Sensors

File: sensors.kicad_sch

Display

File: display.kicad_sch

Rev A room node: ESP32-S3-WROOM-1-N16R8, USB-C 5 V in, 3.3 V LDO.
All sheets connect through global labels and the +5V / +3V3 / GND power symbols.
Source of truth: docs/pcb/schematic.md and docs/pcb/pinout.md.
Power tree: USB-C VBUS (+5V) -> AP7361C LDO -> +3V3 (module, mic, AHT20, e-paper).
+5V also feeds the MAX98357A amp and the LED strip.

Hierarchy: 6 sheets, connected by global labels

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Sheet: /

File: room-node.kicad_sch

Title: Room node rev A

Size: A4

Date: 2026-10-01

Rev: A

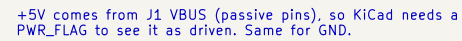
KiCad E.D.A. 10.0.6

Id: 1/7



5. Status LED (GPIO2)

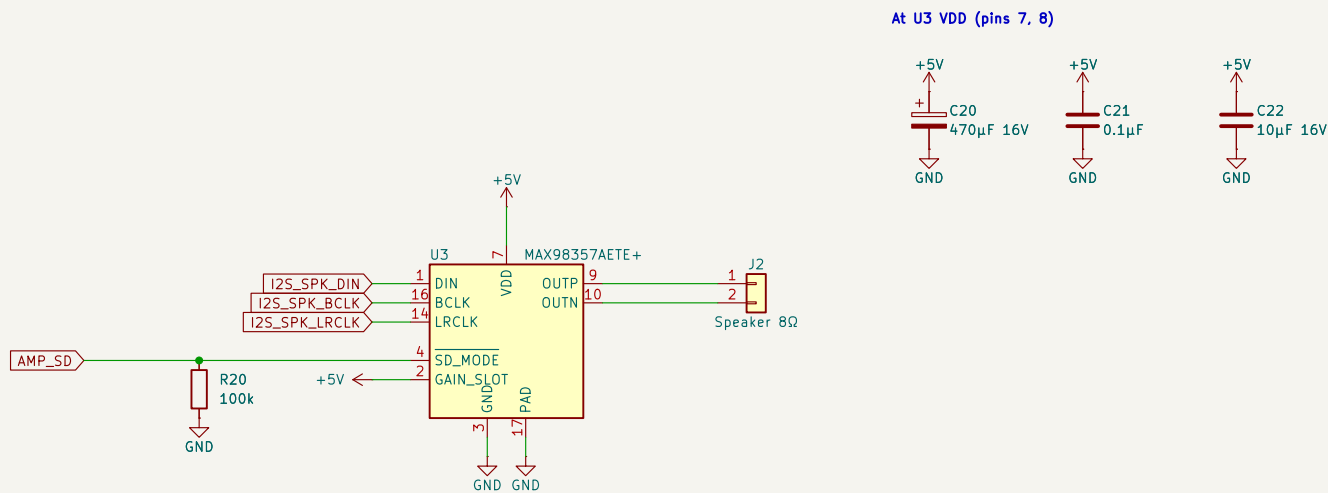
TP1



Title: Room node rev A – MCU, power, USB

[illegible]

I2S class-D amplifier (MAX98357A) -> speaker



GAIN_SLOT to VDD = 6 dB gain (tested on the breadboard; 9 dB clips).
SD_MODE: R20 100k keeps the amp off during boot. GPIO HIGH = on, left channel.
EP (pin 17) is not connected inside the chip; wire it to GND, vias in the pad.
C20 + C21 + star ground removed the breadboard crackle. Keep all three, at the chip.
Speaker: Dayton CE32A-8 (8 Ω). Do not connect either speaker wire to GND.
NC pins 5, 6, 12, 13 are hidden on the stock symbol (no connect).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 2

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Sheet: /Audio Out/
File: audio_out.kicad_sch

Title: Room node rev A – Audio out

Size: A4

Date: 2026-10-01

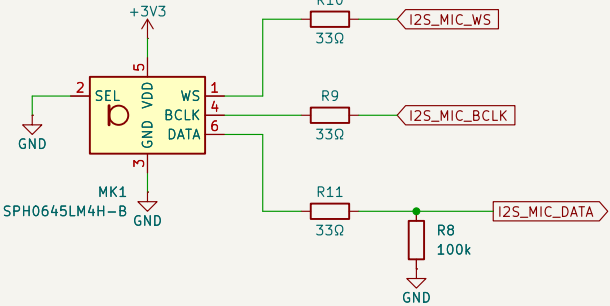
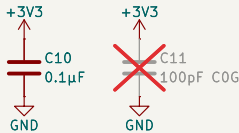
Rev: A

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Id: 3/7

I2S MEMS microphone (SPH0645LM4H-B)

At MK1 VDD (pin 5)



SELECT (pin 2) to GND = left channel.
R9/R10 (33 Ω) at U1, R11 at MK1: optional damping; use 0 Ω if removed.
R8 100k on I2S_MIC_DATA to GND. C11 100 pF C0G is DNP (do not fit).
Use the stock symbol/footprint (Sensor_Audio). Footprint has the sound hole;
GND ring around it. Mic mute is software only (no mute switch).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 3

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Sheet: /Microphone/
File: microphone.kicad_sch

Title: Room node rev A – Microphone

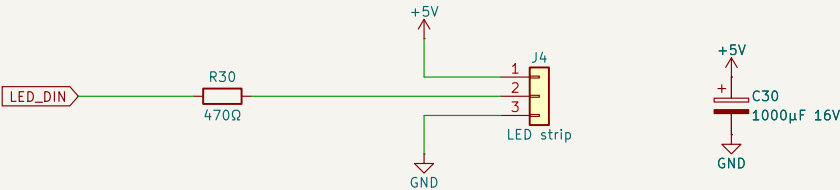
Size: A4 Date: 2026-10-01

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Rev: A

Id: 4/7

WS2812B LED strip connector



J4: 1 = +5V, 2 = DIN, 3 = GND. JST-XH 3 A; full white is ~1.6 A.
R30 near U1. C30 across J4 pins 1 and 3, at J4.
Strip: 27 x WS2812B, off-board. No level shifter (works on the breadboard);
if the strip flickers on rev A, rev B adds a 74AHCT125.

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 4

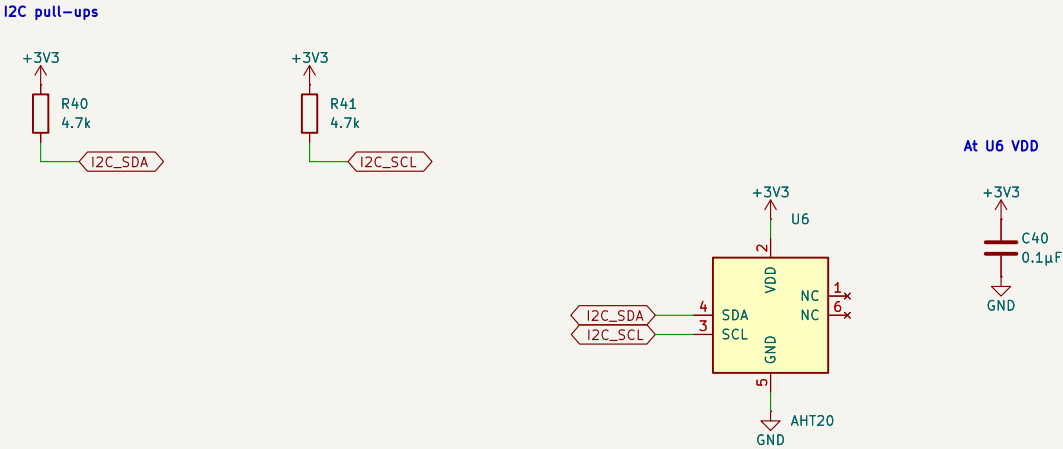
Squarewave Labs

Sheet: /LED Ring/
File: led_ring.kicad_sch

Title: Room node rev A – LED ring

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 5/7

Temperature / humidity (AHT20, I2C 0x38)



Pin 1 and 6 are NC on the AHT20 (no-connect pin type).
Project-library part re-made from LCSC C2757850 (6 pads, 1.0 mm pitch);
pinout per Aosong AHT20 datasheet v1.1 section 3.
Place U6 on a board edge, far from U1, U2, U3 and J4, with slots on three sides
(board heat reads as room temperature).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 5

Squarewave Labs

Sheet: /Sensors/
File: sensors.kicad_sch

Title: Room node rev A – Sensors

Size: A4

Date: 2026-10-01

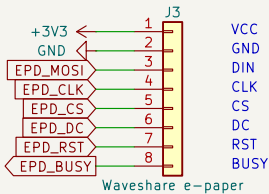
Rev: A

KiCad E.D.A. 10.0.6

Id: 6/7

Waveshare 2.9" e-paper module (with its driver board) via cable

At J3 pin 1



Check the pin order and count on your cable. Newer Waveshare modules have a 9th pin (PWR): if yours has it, use a 9-pin connector and tie PWR to +3V3.
Cable: needs a PH-to-PH 8-pin cable that maps pin 1 to pin 1 (beep it out before power-on). Option: 1x8 2.54 mm header (PinHeader_1x08_P2.54mm_Vertical) for the existing jumper wires. Same pin order either way.

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 6

Squarewave Labs

Sheet: /Display/
File: display.kicad_sch

Title: Room node rev A – Display

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 7/7